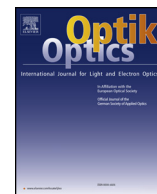




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Original research article

Indoor visible light high precision three-dimensional positioning algorithm based on single LED lamp

Ling Qin, Ben Niu, Baoshan Li, Yongxing Du*

School of Information Engineering, Inner Mongolia University of Science and Technology, Inner Mongolia, Baotou, 014010, China



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ABSTRACT

With the rapid development of information technology, there is a growing demand for people in indoor location services. This paper proposes a novel high-precision indoor three-dimensional positioning algorithm for single light emitting diode (LED). By using three tilted photodetectors (PD) and one horizontal PD as receivers. Calculating the ratio of received optical power of the three tilted PDs to the horizontal PD respectively. According to the relationship of position between the PDs and the tilt angle, azimuth angle, placement angle of the tilted PD and other parameters, establish a mathematical relationship between the received optical power ratio and the receiver's three-dimensional coordinates, calculate the three-dimensional coordinates of the receiver at any position. The simulation results show that in the positioning area of $1 \times 1 \times 1.5 \text{ m}^3$, when the inclination angle α is 25° , the azimuth angle difference θ is 90° , the placement angle $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, the distance r between the horizontal PD center and the tilted PD center is 1 cm, the positioning accuracy is better, the point with the positioning error within 3 cm accounts for 84 % of the total test points, and the average positioning error is 2.52 cm. Compared with the traditional indoor positioning technology, the cost of the positioning technology used in this paper is greatly reduced, the positioning accuracy is significantly improved, and can get three-dimensional coordinates.

1. Introduction

Recently, Global Positioning System (GPS) has been widely used in car navigation, mapping and other fields. When it comes to indoor positioning, the indoor environment is more complicated, the signal from the satellite is blocked by obstacles, the positioning error by GPS in the room will become very big [1]. With the development of science and technology, researchers have proposed many indoor positioning technologies based on wireless electromagnetic waves, such as infrared positioning, wireless local area network (WLAN), radio frequency identification (RFID), Bluetooth, ultra-wideband (UWB), etc. [2]. However, these positioning technologies both have obvious disadvantages: Although infrared positioning technology and UWB positioning technology have higher positioning accuracy, but additional equipment is required, and the cost is high. The price of the Bluetooth device is relatively expensive, and for the complex environment, the stability of the Bluetooth system is poor, and the interference by the noise signal is large. RFID technology has a short working distance, and the positioning error is large, the average positioning error is as high as 1 m. The WIFI transceiver can only cover an area within a radius of 90 m, and is easily interfered by other signals, the average positioning error is as high as 2 ~ 3 m [3]. Using visible light communication (VLC) for indoor positioning is a new way to solve indoor positioning problems [4]. The frequency range of visible light is much wider than that of RF, and there is no electromagnetic interference, which is suitable

* Corresponding author.

E-mail address: dymql@imust.edu.cn (Y. Du).<https://doi.org/10.1016/j.ijleo.2019.163786>

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for hospitals, mines and other places. The visibility of visible light relative to RF signals is better, more convenient and fast, and the positioning accuracy is also high, which has become a hot spot of research on optical communication researchers at this stage [5].

Traditional indoor visible light localization methods are mainly based on Time of arrival (TOA), Time Difference of arrival (TDOA), Angle of arrival (AOA), received signal strength indication (RSSI) method, sensor and imaging method [6]. Among them, although the TOA method is simple in principle, it requires strict synchronization between the transmitting end and the receiving end clock, and requires high hardware. Similarly, the TDOA and AOA methods have higher hardware requirements [7]. Image sensor imaging involves image processing and has a long positioning time [8].

The visible light positioning based on LED lights can also be classified according to the number of light sources, one is based on multiple LED light sources for position determination, the other is to perform indoor positioning according to a single LED [9]. Nowadays, domestic and foreign research on multi-lamp positioning and single-lamp positioning has been done [10]. For multi-lamp positioning technology. In 2016, Yin, F. X. et al. proposed an indoor visible light reverse positioning system. The experimental results show that the positioning error of the system is less than 0.2 m [11]. In 2018, Li, N et al. proposed an indoor positioning system based on visible light communication technology. The simulation results show that the centimeter-level positioning accuracy can be obtained at 5 heights in the $5 \times 5 \times 4 \text{ m}^3$ space. [12]. In 2018, Wang, Q et al. proposed an indoor visible light localization system based on gesture recognition, which achieved decimeter-level positioning accuracy. [13]. In 2019, Chuang, Y. et al. proposed a localization algorithm combining repetition unit and machine learning: The second-order regression ML algorithm and the polynomial three-edge ML algorithm were discussed respectively, and the centimeter-level positioning accuracy was obtained [14]. For single-lamp positioning technology, in 2017, Zhang, R et al. proposed a visible light positioning system based on a single LED lamp, which achieved centimeter-level positioning accuracy in the range of $3 \times 3 \text{ m}^2$, and solved the problem of sparse layout of indoor LED lights and the positioning accuracy caused by insufficient receiver field of view. [15]. In 2018, Yu, X. H. et al. proposed a visible light positioning system using a single LED lamp and a new type of receiver. The receiver consists of a horizontal photodetector (PD) and two tilted PDs. The experimental results show that this system can avoid inter-cell interference caused by different light sources, and it's suitable for scenes with limited number of LEDs [16]. In 2019, Ji, Y. Q. et al. proposed a single LED lamp indoor positioning system based on CMOS image sensor. This system can achieve centimeter-level positioning accuracy regardless of the horizontal or oblique position of the mobile phone [17]. In summary, the indoor positioning of a single LED lamp has the following advantages compared to multi-lamp positioning:

- 1) Solving the problem that the positioning accuracy of some indoor LED lights is sparse and the accuracy of the receiver is insufficient.
- 2) Solving the problem of large positioning error caused by inter-symbol interference between lamps and lamps in multi-lamp positioning [18].
- 3) The positioning speed is fast and the algorithm is relatively simple.

The single LED lamp positioning algorithm proposed in this paper can not only achieve high-precision three-dimensional positioning, but also significantly improve the positioning accuracy and greatly reduce the equipment cost compared with the traditional wireless electromagnetic wave indoor positioning technology. Compared with the traditional multi-lamp positioning technology, the proposed algorithm can effectively solve the problem of large positioning error caused by the interference between symbols of multiple light sources, and does not need to distinguish the optical signals emitted by multiple light sources, which reduces the complexity of the equipment and makes the positioning time faster.

2. Principle of location algorithms

The indoor three-dimensional positioning schematic is shown in Fig. 1(a). Assuming that the LED obeys the Lambert radiation model, the channel gain [19] under the direct line-of-sight link can be expressed as:

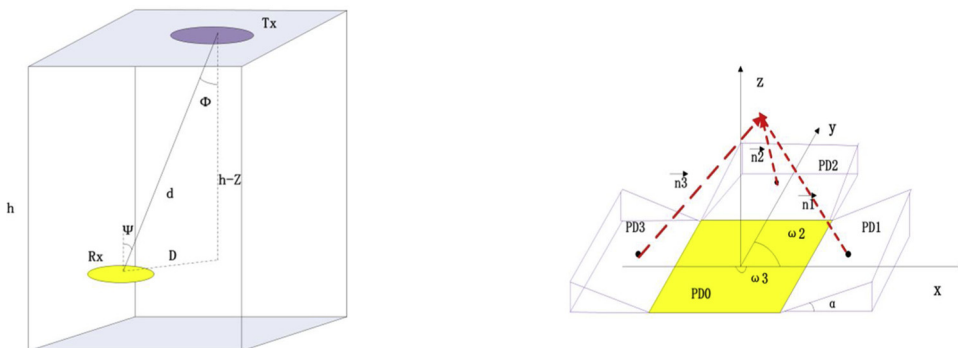


Fig. 1. (a) 3D visible light indoor positioning model (b) Receiver model schematic.

$$H = \begin{cases} \frac{(m+1)}{2\pi d^2} A \cos^m(\varphi) T_s(\psi) g(\psi) \cos(\psi) & 0 \leq \psi \leq \psi_c \\ 0 & \psi > \psi_c \end{cases} \quad (1)$$

In Eq. (1), A is the effective receiving area of the receiver, d is the distance between the receiver and the LED light source, (ψ) is the incident angle of the receiver, ϕ is the emission angle of the LED, $T_s(\psi)$ is the gain of optical filter, $g(\psi)$ is the gain of optical concentrator, ψ_c is the field of view angle of the receiver and m is the Lambert order of the light source, m can be derived from the following formula:

$$m = \frac{\ln 2}{\ln \cos \phi_{1/2}} \quad (2)$$

In Eq. (2), $\phi_{1/2}$ is the half power angle of the LED light source, that is, the radiation power of the LED light source at this angle is 1/2 of the central power [20].

Fig. 1(b) shows the receiver model, which consists of a horizontal PD (denoted as PD₀) and three tilted PDs (denoted as PD_i, $i = 1, 2, 3$). In the present positioning system, according to the Eq. (1), if the transmission power of the LED is known to be P_t , the received power of each PD can be expressed as:

$$P_{ri} = P_t \cdot H \quad (i=0,1,2,3) \quad (3)$$

The positional relationship between PD₀ and each tilted PD is as follows:

$$\begin{aligned} X_{ri} &= X_r + r \cos \alpha \cos(\omega_i) \\ Y_{ri} &= Y_r + r \cos \alpha \sin(\omega_i) \\ Z_{ri} &= Z_r + r \sin(\alpha) \end{aligned} \quad (4)$$

Where (X_r, Y_r, Z_r) and (X_{ri}, Y_{ri}, Z_{ri}) are the coordinates of the horizontal PD and the oblique PD, respectively, r is the distance between the center and the horizontal PD center, α is the tilt angle of the tilted PD. In the receiver herein, the tilt angles α of the three tilted PDs are equal, and the horizontal PD center and the tilted PD center distance r is also equal. ω is the angle between the line between the tilted PD center and the horizontal PD center and the positive direction of the x-axis, which can be understood as the tilt angle of the tilted PD.

In this paper, the distance between horizontal and inclined PDs is very close, so the transmission distance from LED to horizontal PDs is approximately equal to that from inclined PDs. Therefore, at any point in the three-dimensional space, the emission angle of the LED is always meet. Let ψ_0 and ψ_i be the incident angles of horizontal and inclined PD respectively. The relationship between incident angles is as follows:

$$\begin{aligned} \cos(\psi_0) &= \cos(\phi) = \frac{h - Z_r}{d} \\ \cos(\psi_i) &= \frac{(X_0 - X_{ri}) \sin(\alpha) \cos(\beta_i) + (Y_0 - Y_{ri}) \sin(\alpha) \sin(\beta_i) + (Z_0 - Z_{ri}) \cos(\alpha)}{\sqrt{(X_0 - X_{ri})^2 + (Y_0 - Y_{ri})^2 + (Z_0 - Z_{ri})^2}} \end{aligned} \quad (5)$$

In Eq. (5), β_i is the azimuth angle of the inclined PD, it is expressed as the angle between the projection of the normal vector of the inclined PD receiving surface to the xoy plane and the positive direction of the x-axis. The range of β_i is $0^\circ \sim 360^\circ$. (X_0, Y_0, Z_0) is the coordinates of the LED light source. In this paper, $\alpha, \beta_i, \omega_i, r$ and the coordinates of the LED light source are all known.

When the incident angle of each PD and the positional relationship of each PD are determined, three-dimensional positioning can be performed. In the process of three-dimensional positioning, the coordinates of the receiver at any position are set to (X_r, Y_r, Z_r) . Substituting Eq. (5) into Eq. (1), it can be calculated that the received optical power of the horizontal PD and the oblique PD can be expressed as:

$$P_{r0} = \frac{k(h - Z_r)^{m+1}}{d^{m+3}} \quad (6)$$

$$P_{ri} = \frac{k(h - Z_r)^m}{d^{m+3}} [(X_0 - X_{ri}) \sin(\alpha) \cos(\beta_i) + (Y_0 - Y_{ri}) \sin(\alpha) \sin(\beta_i) + (Z_0 - Z_{ri}) \cos(\alpha)] \quad (7)$$

In Eq. (6) and Eq. (7), k is a constant. Because in the positioning process, we use a single LED light, and the same PD is used. According to the received optical power of each PD, using Eq. (6) and Eq. (7), the ratio of the tilted PD received optical power to the horizontal PD received optical power can be determined as:

$$RSSR_i = \frac{1}{(h - Z_r)} [(X_0 - X_{ri}) \sin(\alpha) \cos(\beta_i) + (Y_0 - Y_{ri}) \sin(\alpha) \sin(\beta_i) + (Z_0 - Z_{ri}) \cos(\alpha)] \quad (8)$$

In Eq. (8), $RSSR_i = P_{ri}/P_{r0}$ ($i = 1, 2, 3$), it is expressed as the ratio of the received optical power of each oblique PD to the horizontal PD received optical power. Substituting Eq. (4) into Eq. (8), the following relationship can be obtained:

$$RSSR_i = \frac{1}{(h - Z_r)} [(X_0 - (X_r + r\cos(\alpha)\cos(\beta_i)))\sin(\alpha)\cos(\beta_i) + (Y_0 - (Y_r + r\cos(\alpha)\sin(\omega_i)))\sin(\alpha)\sin(\beta_i) + (Z_0 - (Z_r + r\sin(\alpha)))\cos(\alpha)] \quad (9)$$

In Eq. (9), let:

$$\begin{aligned} A_i &= \sin(\alpha)\cos(\beta_i) \\ B_i &= \sin(\alpha)\sin(\beta_i) \\ T_i &= [(X_0 - r\cos(\alpha)\cos(\omega_i))]A_i + [Y_0 - r\cos(\alpha)\sin(\omega_i)]B_i + (Z_0 - r\sin(\alpha))\cos(\alpha) \end{aligned} \quad (10)$$

We can get Eq. (11) :

$$T_i - hRSSR_i = X_r A_i + Y_r B_i + Z_r (\cos(\alpha) - RSSR_i) \quad (11)$$

It can be seen from Eq. (11), the three-dimensional coordinates of the horizontal PD are only related to the value of the received optical power ratio $RSSR_i$. Since the three-dimensional coordinates of the LED lamp are known, the positional relationship between the respective PDs and the angle parameter of the tilt PD are known, so according to the optical power ratio of the three tilted PDs to the horizontal PD and the angle parameters of the three tilted PDs, The three-dimensional coordinates of the horizontal PD are obtained by solving the ternary equations by Eq. (11). The specific solution process is as follows:

let :

$$\begin{aligned} M_1 &= \cos(\alpha) - RSSR_1 \\ M_2 &= \cos(\alpha) - RSSR_2 \\ M_3 &= \cos(\alpha) - RSSR_3 \\ N_1 &= T_1 - hRSSR_1 \\ N_2 &= T_2 - hRSSR_2 \\ N_3 &= T_3 - hRSSR_3 \end{aligned} \quad (12)$$

Substituting Eq. (12) into Eq. (11), the following equations can be obtained:

$$\begin{cases} A_1 X_r + B_1 Y_r + M_1 Z_r = N_1 \\ A_2 X_r + B_2 Y_r + M_2 Z_r = N_2 \\ A_3 X_r + B_3 Y_r + M_3 Z_r = N_3 \end{cases} \quad (13)$$

let :

$$\begin{aligned} P_1 &= \frac{M_2 A_1 - M_1 A_2}{M_1} \\ P_2 &= \frac{M_2 B_1 - M_1 B_2}{M_1} \\ P_3 &= \frac{M_2 N_1 - M_1 N_2}{M_1} \\ K_1 &= \frac{A_3 M_1 - A_1 M_3}{M_1} \\ K_2 &= \frac{B_3 M_1 - B_1 M_3}{M_1} \\ K_3 &= \frac{M_1 N_3 - M_3 N_1}{M_1} \end{aligned} \quad (14)$$

Bringing Eq. (14) into Eq. (13), the three-dimensional coordinates of the horizontal PD can be solved as:

$$\begin{cases} X_r = \frac{K_2 P_3 - K_3 P_2}{P_1 K_2 - P_2 K_1} \\ Y_r = \frac{P_3 K_1 - K_3 P_1}{P_2 K_1 - K_2 P_1} \\ Z_r = \frac{N_1 - A_1 X_r - B_1 Y_r}{M_1} \end{cases} \quad (15)$$

According to the Eq. (15), when the ratio of the received optical powers of the three tilted PDs to the horizontal PD is known, the three-dimensional coordinates of horizontal PD can be calculated at an arbitrary position according to the relationship of position between the respective PDs and the angle parameter of the tilted PD. According to the receiver model that is shown in Fig. 1(b), the three-dimensional coordinates of the horizontal PD are the three-dimensional coordinates of the receiver. By substituting the Eq. (15) into the Eq. (4), the three-dimensional coordinates of the three oblique PDs at arbitrary positions can be obtained. Let the actual coordinates of the receiver are $(X_{\text{real}}, Y_{\text{real}}, Z_{\text{real}})$, the positioning coordinates are (X_r, Y_r, Z_r) , the positioning errors are P_e . The positioning error of the receiver in each test point in the three-dimensional space can be expressed as:

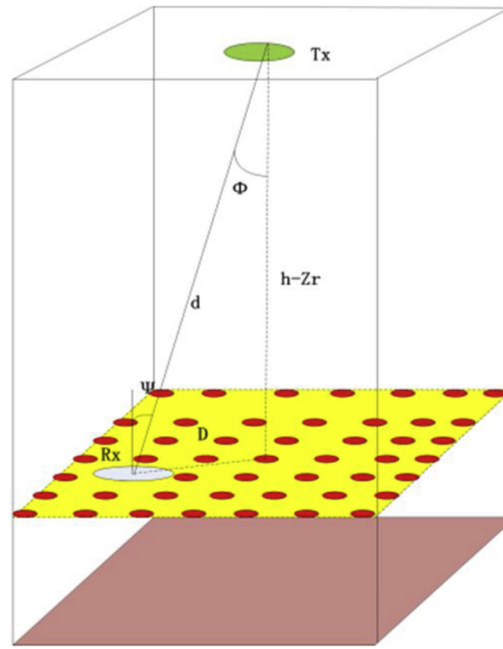


Fig. 2. experimental system model.

$$P_e = \sqrt{(X_r - X_{real})^2 + (Y_r - Y_{real})^2 + (Z_r - Z_{real})^2} \quad (16)$$

3. Simulation of positioning system

In order to evaluate the performance of the positioning system, an experimental system is constructed in this paper. As is shown in Fig. 2, the length, width and height of the initial positioning area are set to 1 m, 1 m, and 1.5 m respectively, and the coordinates of the LED light source are (0.5, 0.5, 1.5). The interval between adjacent test heights is 0.2 m. on each test height, 7×7 test points are set. The initial value of the distance r between the horizontal PD center and the inclined PD center is set to 1 cm, the initial value of the azimuth difference θ of the adjacent two oblique PDs is set to 90° , the placement angles ω_1 , ω_2 , and ω_3 of the three tilted PDs are set to 0° , 90° , and 180° respectively. This paper discusses the factors about what affecting the positioning error: including the tilt angle α of the tilt PD, the azimuth angle difference θ of the adjacent two tilted PDs, the placement angle ω_i of the three tilted PDs ($i = 1, 2, 3$), and the distance r between the center of horizontal PD tilted PD. So these influencing factors will be discussed separately below:

Firstly, when θ , r , ω_i remain unchanged, only the inclination angle α of inclined PD is changed. The simulation results show that when the inclination angle is greater than 30 degrees and the receiver is measured at 1.2 m height, the receiver can not locate at some edge test points. This is because when the inclined angle of inclined PD is greater than 30 degrees, the incident angle of inclined PD at some test edge points is larger than the field of view of inclined PD. According to formula (1), the optical power received by inclined PD at some test edge points is zero. So in the course of positioning, the inclination angle of inclined PD should be less than 30 degrees.

In order to describe more clearly of the variation of positioning error with inclination angle α , Fig. 3 shows the distribution of positioning error of receiver at 0.4 m altitude when inclination angle α varies at 10, 15, 20 and 25 degrees.

It can be seen from Fig. 3 that when the test height of the receiver is constant, as the tilt angle α increases, the positioning error of the edge test point gradually decreases. Table 1 shows the positioning error when the tilt angle α is changed at 10° , 15° , 20° , 25° and when the test height of the receiver is 0.4 m.

According to Table 1, when other factors remain unchanged and the receiver measures at the same altitude, the bigger inclination angle of inclined PD is, the smaller average positioning error is.

When α is constant, the positioning errors of the receiver are also different at different test altitudes. Fig. 4 shows the distribution of positioning errors of the receiver at 0.2 m, 0.4 m, 0.6 m and 0.8 m altitudes when α equals to 15° .

It can be seen from Fig. 4, when the tilt angle of the tilted PD is constant, the positioning error of the edge test points increases slightly with the increase of the test height of the receiver. The average positioning errors of the receiver at 0.2 m, 0.4 m, 0.6 m and 0.8 m altitudes are shown in Table 2 when α is 15° .

It can be seen from Table 2, when the tilt angle is constant, the average positioning error increases slightly with the increase of the test height of the receiver. Analysis of Tables 1 and 2 leads to conclusions: when other factors remain unchanged, the bigger the α is, the smaller the average positioning error is. When the tilt angle α is fixed, the average positioning error increases slightly with the increase of the receiver's test height.

The influence of azimuth angle difference θ between adjacent inclined PDs on positioning error is also studied. The distance r

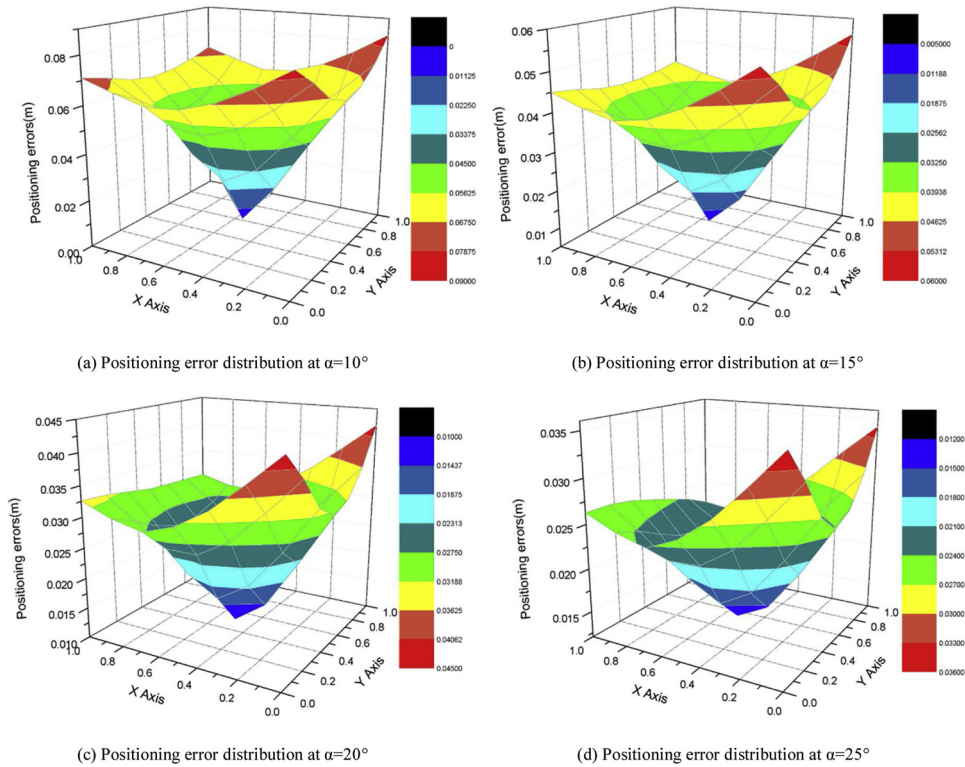


Fig. 3. Position error distribution of the receiver at a height of 0.4 m when the tilt angle α is changed at 10° , 15° , 20° , and 25° . (a) Positioning error distribution at $\alpha = 10^\circ$ (b) Positioning error distribution at $\alpha = 15^\circ$. (c) Positioning error distribution at $\alpha = 20^\circ$ (d) Positioning error distribution at $\alpha = 25^\circ$.

Table 1

The positioning errors of inclination angle α at 10° , 15° , 20° , 25° and test height is 0.4 m.

Inclination angle $\alpha(^{\circ})$	Maximum positioning error (cm)	Minimum positioning error (cm)	Mean positioning error (cm)
10	8.31	0.51	5.56
15	5.65	0.77	3.67
20	4.31	1.02	3.06
25	3.52	1.26	2.31

between horizontal and inclined PDs is unchanged. The placing angle of inclined PDs is set at $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$. The relationship between the average positioning error and the azimuth angle difference is shown in Fig. 5 when the inclination angle is 10° , 15° , 20° and 25° , respectively.

It can be seen from Fig. 5, when the inclination angle of inclined PD is constant, with the increase of θ , the average positioning error decreases first, then becomes stable and finally increases. When $\theta < 90^\circ$, the positioning error decreases gradually with the increase of θ . when $90^\circ < \theta < 135^\circ$, the positioning error tends to be stable with the change of θ . When $\theta > 135^\circ$, the positioning error suddenly increases. It can be seen that θ has a great influence on the positioning error.

Changing the placement angle ω of inclined PDs. Set $\omega_1 = 0^\circ$, $\omega_2 = 120^\circ$, $\omega_3 = 240^\circ$. When the inclination angle is 15° , The variation curves of average positioning error with azimuth difference θ are compared with those with azimuth difference θ when $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, and inclination angle α is 15° . The comparison of the variation curves of average positioning error with azimuth difference θ is shown in Fig. 6.

It can be seen from Fig. 6 that when α is constant, when $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, the positioning effect is better than $\omega_1 = 0^\circ$, $\omega_2 = 120^\circ$, and $\omega_3 = 240^\circ$. When studying the influence of other influencing factors on the positioning error, the placement angle of the tilt PD is always set to $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$.

When the center distance r between the horizontal PD and the tilt PD is changed, the positioning error is also affected. When r is 1 cm, 5 cm and 10 cm respectively, the average positioning error is shown in Table 3.

It can be seen from Table 3 that with other influencing factors unchanged, the larger the distance r between the horizontal PD center and the inclined PD center is, the larger the positioning error is. According to the above analysis, we know that the α of the tilt PD, the azimuth angle difference θ of the adjacent two tilt PDs, the placement angle ω of the tilt PD, and the distance r between the center of the horizontal PD and the center of the tilt PD will all have impact on positioning errors. According to the above analysis of

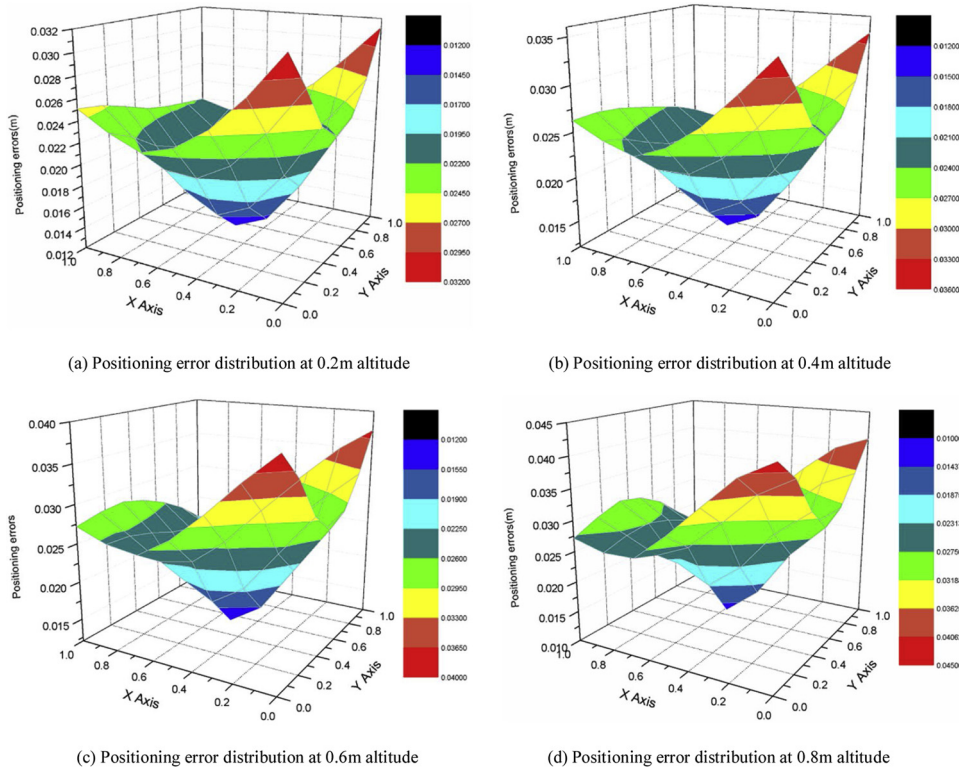


Fig. 4. Distribution of positioning errors when the inclination angle is 15° and the receiver is measured at altitudes of 0.2 m, 0.4 m, 0.6 m and 0.8 m. (a) Positioning error distribution at 0.2m altitude (b) Positioning error distribution at 0.4m altitude. (c) Positioning error distribution at 0.6m altitude (d) Positioning error distribution at 0.8m altitude.

Table 2

α is 15° , average positioning errors of receivers at altitudes of 0 m, 0.4 m, 0.8 m and 1.2 m.

Height of receiver (m)	0.2	0.4	0.6	0.8
Average positioning error (cm)	3.29	3.67	4.13	4.62

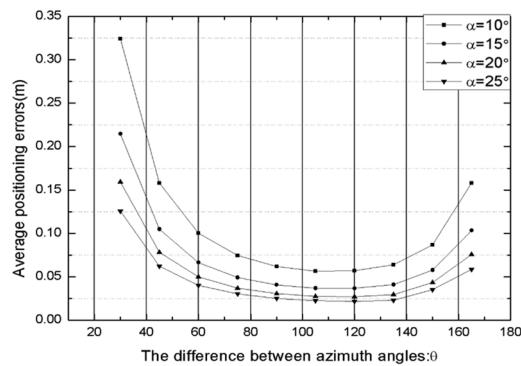


Fig. 5. The variation curve of average positioning error with azimuth difference θ when $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$.

the influencing factors of the positioning error, when $\alpha = 25^\circ$ and the azimuth angle difference θ of the adjacent two tilting PDs is constant at 90° , the placement angles of the three tilting PDs are respectively $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, the distance r between the center of the horizontal PD and the center of the tilted PD is 1 cm, the positioning error is the smallest. Next, this paper will analyze the positioning situation when $\alpha = 25^\circ$, the azimuth angle difference $\theta = 90^\circ$, $r = 1$ cm, $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$.

When $\alpha = 25^\circ$, $\theta = 90^\circ$, $r = 1$ cm, $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, the positioning points of the receiver at the test altitudes of 0.2 m, 0.4 m, 0.6 m and 0.8 m are compared with the actual position coordinates is shown in Fig. 7 below.

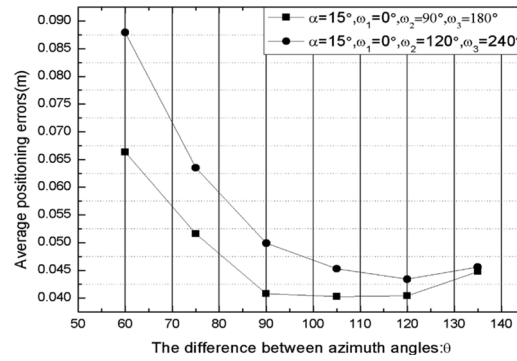


Fig. 6. $\alpha = 15^\circ$, the average positioning error varies with the azimuth angle θ when the placement angle changes.

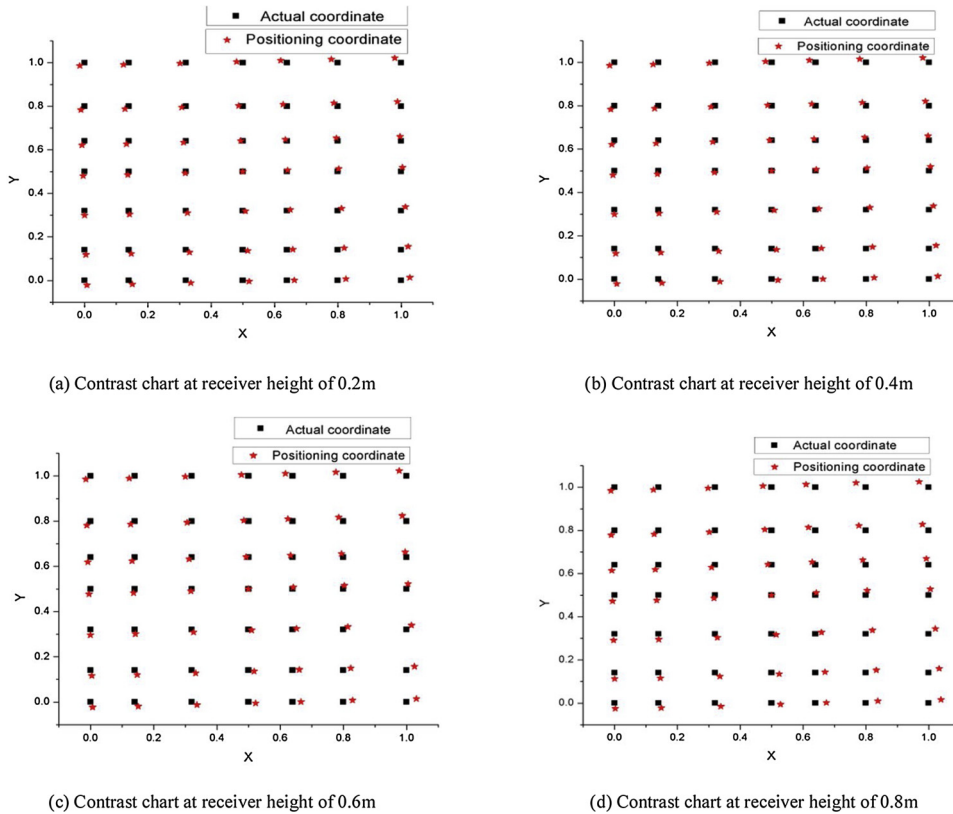


Fig. 7. Comparison of positioning coordinates and actual coordinates of receiver at different altitudes.

(a) Contrast chart at receiver height of 0.2 m (b) Contrast chart at receiver height of 0.4 m. (c) Contrast chart at receiver height of 0.6 m (d) Contrast chart at receiver height of 0.8 m.

Table 3

The variation of average positioning error with the distance r .

Receiver center distance (cm)	1	5	10
Average positioning error(cm)	2.52	13.10	27.53

It can be seen from the above figure that in the case of $\alpha = 25^\circ$, $\theta = 90^\circ$, $r = 1$ cm, $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, the positioning error is small, on each test height the upper positioning coordinates are not much different from the actual coordinates. It can be seen by observing the distance between the black box and the red five-pointed star. As the height of the receiver test increases, the positioning error increases slightly. When $\alpha = 25^\circ$, $\theta = 90^\circ$, $r = 1$ cm, $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, comparison of the proposed algorithm error location histogram and the literature [16] positioning error histograms is show in Fig. 8 below:

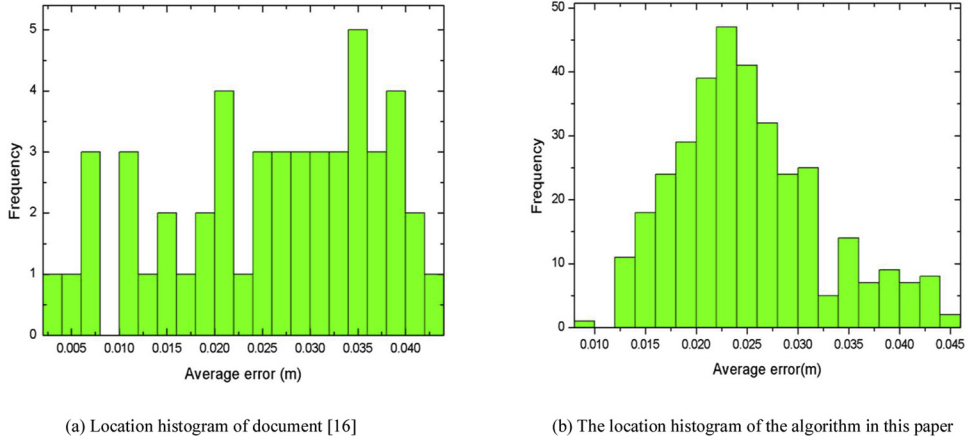


Fig. 8. comparison of the proposed algorithm error location histogram and the literature [16] positioning error histograms.
(a) Location histogram of document [16] (b) The location histogram of the algorithm in this paper.

It can be seen from Fig. 8 that for the positioning algorithm proposed in [16], the positioning error is mainly concentrated at 3.5 cm, and the point with the positioning error within 3 cm accounts for 63 % of the total test points, and the algorithm positioning error proposed in this paper is mainly concentrated on 2.25 cm. Points with a positioning error within 3 cm accounted for 84 % of the total test point. It can be seen that compared with the positioning algorithm in [16], the positioning algorithm proposed in this paper can not only achieve high-precision three-dimensional positioning indoors, but also better positioning effect.

When $\alpha = 25^\circ$, $\theta = 90^\circ$, $r = 1$ cm, $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, the cumulative probability distribution of the positioning error is shown in Fig. 9 below:

From the error cumulative distribution map, we can see that the maximum positioning error is 4.49 cm and the minimum value is 1.24 cm. The simulation results show that the point with the smallest positioning error is directly below the LED, that is, at the center point of each test height plane.

4. Conclusion

This paper proposes a novel indoor three-dimensional positioning algorithm for single LED lamps, using a single LED lamp as the transmitter, three tilt PDs and one horizontal PD as the receiver to receive the light intensity signal, and each tilted PD receiving optical power and horizontal PD receiving. Compared with the power light, the received optical power ratio is substituted into the positioning algorithm, and the three-dimensional coordinates of the receiver are calculated to complete the three-dimensional positioning.

In this paper, the main factors affecting the positioning error are studied, including the tilt angle α of the tilt receiver, the azimuth angle different θ of the tilt receiver, the distance r between the center of the horizontal PD and the tilt PD center, and the placing angle ω of the tilt PD. The simulation results show that when the other influencing factors are constant, when the tilt angle α is within a certain range of variation, the positioning error decreases as the tilt angle α increases. When α is constant, the receiver is tested. As the height increases, the positioning error increases slightly. When only changing the azimuth difference θ of the tilt receiver, when $\theta < 90^\circ$, the positioning error decreases gradually with the increase of θ . When $90^\circ < \theta < 135^\circ$, the positioning error tends to be stable with the change of θ . When $\theta > 135^\circ$, the positioning error suddenly increases. When only the distance r between the horizontal PD center and the inclined PD center is changed, the positioning error increases remarkably as r increases. After analysis, the

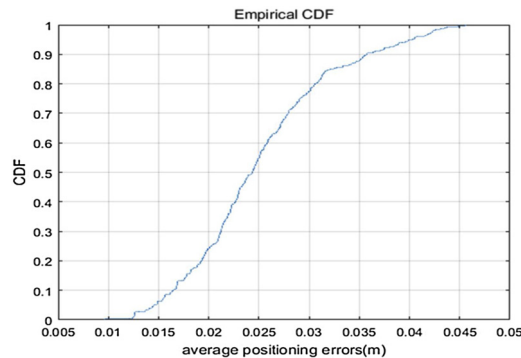


Fig. 9. cumulative error distribution map.

factors affecting the positioning error are mainly the tilt PD tilt angle α , the azimuth angle difference θ of the tilt PD, and the horizontal PD center and the tilt PD center distance r . By analyzing the influencing factors of the positioning error, when the tilt angle $\alpha = 25^\circ$, the azimuth angle difference $\theta = 90^\circ$, $r = 1$ cm, $\omega_1 = 0^\circ$, $\omega_2 = 90^\circ$, $\omega_3 = 180^\circ$, the positioning error is the smallest. The minimum error is 1.24 cm, the maximum error is 4.49 cm, the point with positioning error within 3 cm accounts for 84 % of the total test point, and the average positioning error is 2.52 cm. In this paper, the error distribution histogram and the cumulative probability distribution map are also used to analyze the positioning conditions when the tilt angle $\alpha = 25^\circ$, the azimuth angle difference $\theta = 90^\circ$, $r = 1$ cm, compared with the positioning algorithm in [16]. The positioning algorithm can not only achieve high-precision three-dimensional positioning indoors, but also better positioning effect.

The single LED lamp three-dimensional positioning algorithm proposed in this paper has low positioning error. The positioning algorithm is simple and novel compared with the traditional indoor positioning algorithm. It can effectively solve the problem of poor positioning accuracy caused by interference between multiple light source symbols in multi-light source positioning, and it does not need to be more. The optical signals emitted by the light sources are distinguished, which reduces the complexity of the device and the positioning time is shorter.

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